

Bank Telemarketing Campaign Strategy

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Business Understanding: The Efficiency Gap

Key Challenges

- 💰 **High Operational Costs:** Significant agent time per call.
- Low Conversion:** Bulk calling leads to wasted resources.
- 👥 **Lack of Prioritization:** Treating all customers as equals.



Data Scope: 45,211 Records (2008-2010)

Category	Key Variables	Description
Customer Data	Age, Job, Marital, Education	Demographic profiling attributes.
Campaign Data	Contact, Month, Day, Duration	Specifics of the last contact attempt.
Financial Health	Balance, Housing, Personal Loan	Customer's economic status and liabilities.
History	Campaign, Pdays, Previous, Poutcome	Previous interaction outcomes.

Data Preparation: Ensuring Integrity



Filtering

Excluded 'unknown' contact channels to focus on actionable Cellular and Telephone segments.



Bucketing

Grouped ages into 10-year intervals (e.g., 28-37) to find demographic sweet spots.



Relabeling

Transformed 'unknown' previous outcomes to 'No Prior Contact' for better historical analysis.

Campaign Efficiency Evaluation

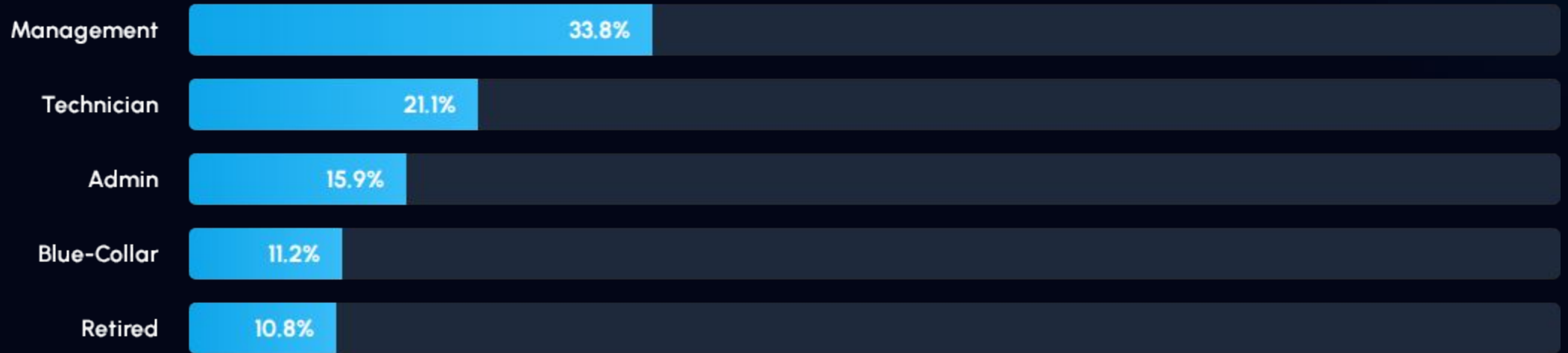
14.8%
Total Conversion Rate

The 85% Opportunity

Out of 32,191 calls, 4,759 subscribed. This indicates that **85.2% of calls were non-productive**.

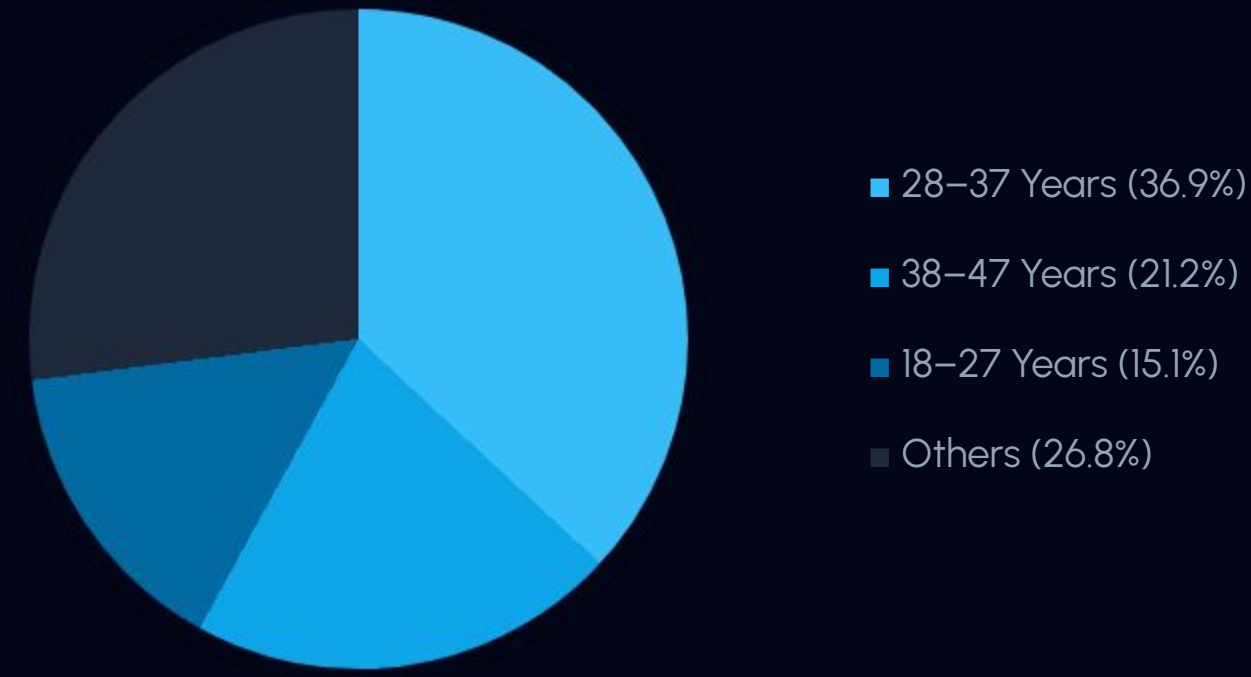
Strategic targeting can significantly reduce agent fatigue and operational waste by focusing on high-probability segments.

Top 5 Segments by Job Profile



Management and Technical staff constitute over **54% of all successful subscriptions.**

Primary Target Age Groups



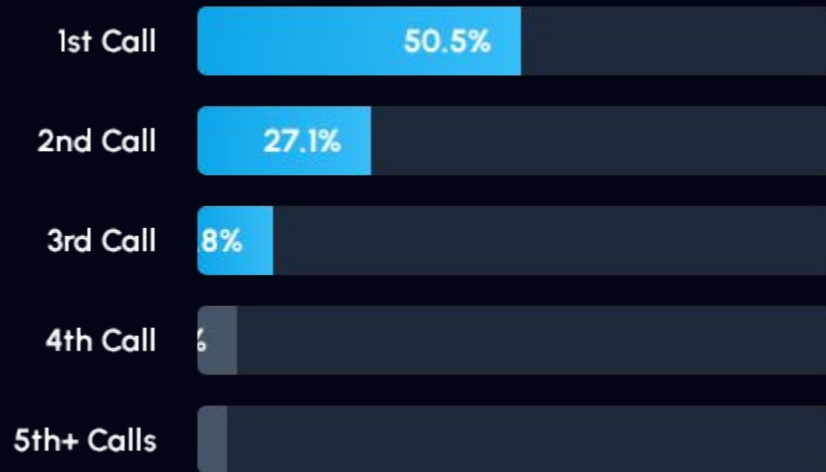
The **28-37 age bucket** is our most dominant and profitable segment.

Monthly Conversion Efficiency



While **July/August** see the highest volume, **February and March** show the best conversion ratios.

The Law of Diminishing Returns



Strategy: Stop at Call #3

Conversion drops by **75%** between the first and third contact.

Maintaining persistence beyond the 3rd attempt is mathematically unproductive. Reallocating these efforts to new leads can increase total capacity by over 20%.

Strategic Action Items

1The "3-Call" Rule

Cap follow-ups at 3 contacts. Transition non-responsive leads to low-cost digital nurture instead of expensive manual calls.

2Cellular Priority

Prioritize **Cellular** contacts (14.9% conv) over landlines. Mobile availability correlates with modern, tech-savvy profiles.

3Historical Success

Re-target customers with '**poutcome = success**'. They convert at 17.78%, significantly higher than the cold lead average.



Implementation Roadmap



Seasonal Focus

Increase budget and staffing during Feb–Mar to capture higher efficiency windows.



Data Quality

Invest in verifying 'unknown' channels. Missing data directly inhibits targeting accuracy.



Model Refinement

Continue moving from descriptive patterns to predictive machine learning scoring.